like a particle and travels either through the left or through the right slit, preventing interference from happening.

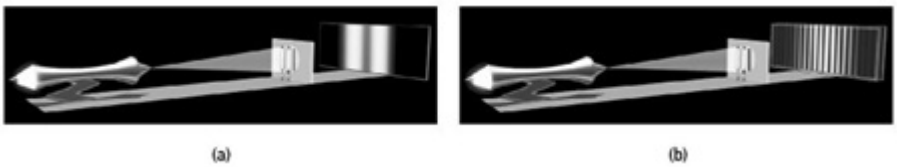


Figure 7.4 In the quantum eraser experiment, equipment placed in front of the two slits marks the photons so that subsequent examination can reveal through which slit each photon passed. In (a) we see that this which-path information spoils the interference pattern. In (b) a device that erases the mark on the photons is inserted just in front of the detector screen. Because the which-path information is eliminated, the interference pattern reappears.

In an experiment carried out by Raymond Chiao, Paul Kwiat, and Aephraim Steinberg, the setup was, schematically, as in Figure 7.4, with a new erasure device inserted just in front of the detection screen. Again, the details are not of the essence, but briefly put, the eraser works by ensuring that regardless of whether a photon from the left slit or the right slit enters, its spin is manipulated to point in one and the same fixed direction. Subsequent examination of its spin therefore yields no information about which slit it passed through, and so the which-path mark has been erased. Remarkably, the photons detected by the screen after this erasure *do* produce an interference pattern. When the eraser is inserted just in front of the detector screen, it undoes—it erases—the effect of tagging the photons way back when they approached the slits. As in the delayed-choice experiment, in principle this kind of erasure could occur billions of years after the influence it is thwarting, in effect undoing the past, even undoing the ancient past.

How are we to make sense of this? Well, keep in mind that the data conform perfectly to the theoretical prediction of quantum mechanics. Scully and Drühl proposed this experiment because their quantum mechanical calculations convinced them it would work. And it does. So, as is usual with quantum mechanics, the puzzle doesn’t pit theory against experiment. It pits theory, confirmed by experiment, against our intuitive sense of time and reality. To ease the tension, notice that were you to